

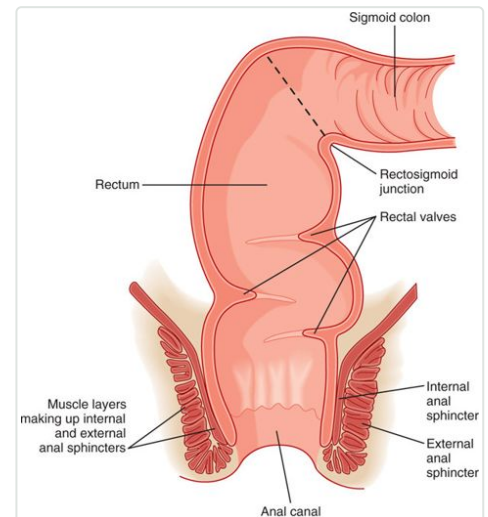
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What keeps the rectum empty most of the time?
2. Which anal sphincter is voluntary, and which nerve supplies it?
3. Which defecation reflex is weak on its own, and what turns it into a powerful one?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Why the rectum is usually empty

- A **weak functional sphincter** sits about **20 cm from the anus**, at the junction between the **sigmoid colon and the rectum**.
- **Sharp angulation** at the same site adds resistance to filling of the rectum.
- A **mass movement** forcing feces into the rectum → **immediate desire to defecate**, with **reflex contraction of the rectum** and **relaxation of the anal sphincters**.
- Between episodes, a **continual dribble** of feces is prevented by **tonic constriction of both anal sphincters** (next section).



Rectosigmoid junction, rectal valves, internal and external anal sphincters.

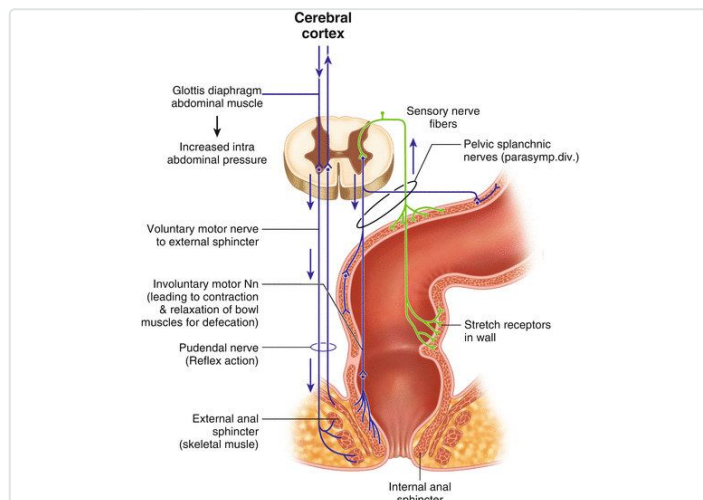
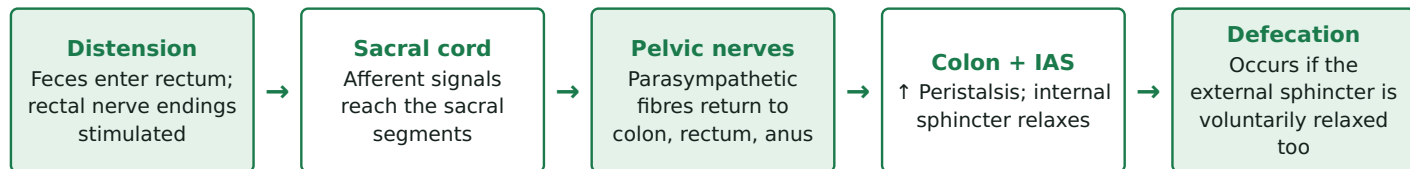
2 Internal vs external anal sphincter

	Internal anal sphincter	External anal sphincter
Muscle	Several-cm thickening of circular smooth muscle inside the anus	Striated muscle that surrounds the internal sphincter and extends distal to it
Control	Involuntary — relaxed by inhibitory signals from the myenteric plexus	Voluntary (conscious or subconscious) via the pudendal nerve , part of the somatic nervous system
Tone	Tonic constriction	Continuously constricted unless conscious signals inhibit it

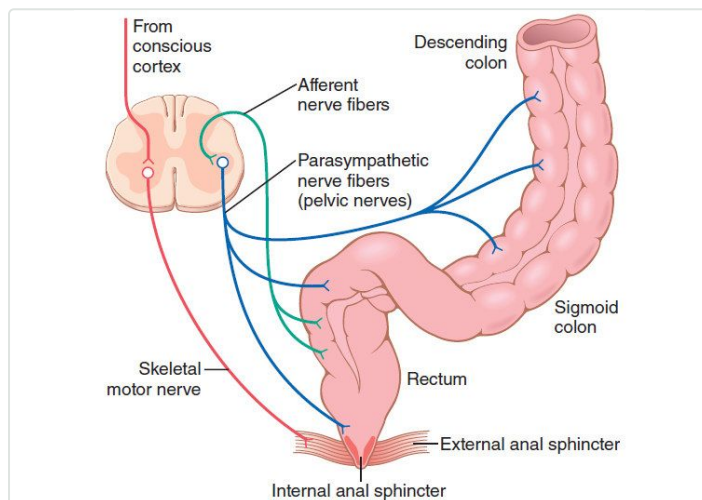
3 The two defecation reflexes

	Intrinsic (myenteric) reflex	Parasympathetic (sacral) reflex
Mediated by	Local enteric nervous system in the rectal wall	Sacral spinal cord segments; parasympathetic fibres in the pelvic nerves
Trigger	Rectal distension → afferent signals spread through the myenteric plexus	Nerve endings in the rectum stimulated → signals to the spinal cord and back
Effect	Peristaltic waves in descending colon, sigmoid and rectum; internal sphincter relaxed by myenteric inhibitory signals	Parasympathetic signals intensify peristaltic waves and relax the internal sphincter (descending colon, sigmoid, rectum, anus)
Strength	Relatively weak when acting alone	Powerful — empties the bowel from the splenic flexure to the anus

VISUAL ANCHOR · THE POWERFUL REFLEX, STEP BY STEP



Full reflex circuit: stretch receptors → pelvic splanchnic nerves (parasympathetic); cortex → voluntary motor nerve / pudendal nerve → external sphincter.



Parasympathetic reflex: afferent fibres to the sacral cord; pelvic-nerve fibres to descending colon, sigmoid, rectum and anus.

4 Assist, control and normal pattern

Cue	Detail
Cord-initiated assist	Deep breath , closure of the glottis , contraction of the abdominal wall (forces colonic contents downward); the pelvic floor relaxes downward and pulls outward on the anal ring to evaginate the feces
Voluntary activation	When convenient: deep breath (diaphragm down) + abdominal contraction → ↑ abdominal pressure → feces forced into the rectum → new reflexes. Never as effective as natural reflexes , so people who often inhibit them are likely to become constipated
Newborns, cord transection	Reflexes cause automatic emptying — no conscious control of the external sphincter
Normal range	2-3 per day to 3-4 per week ; depends on diet, health and stress

5 Other autonomic reflexes that affect bowel activity

Reflex	Trigger	Effect
Peritoneointestinal	Irritation of the peritoneum	Inhibits excitatory enteric nerves → intestinal paralysis , especially in peritonitis
Renointestinal	Irritation of the kidney	Inhibits intestinal activity
Vesicointestinal	Irritation of the bladder	Inhibits intestinal activity

Also named in the lecture (not explained): duodenocolic, gastrocolic, gastroileal and enterogastric reflexes.

6 Diarrhoea vs constipation

	Diarrhoea	Constipation
Definition	↑ frequency, volume and fluid content of faeces	Infrequent or difficult defecation
Motility	Increased motility + decreased absorption	Decreased motility
Mechanism	Contents pass too quickly → not enough time for absorption	Faeces stay long in the colon → excess water absorbed → dry, hard stool
Causes	Lactose intolerance, stress, microbes irritating the GI mucosa	Delaying defecation, colon spasms, low fibre, low fluids, little exercise, emotional stress, certain drugs
Consequence / fix	Dehydration and electrolyte imbalance	Mild laxatives work but are habit-forming ; safer: more fibre, exercise and fluids

P Must-remember pearls

- The rectum is usually empty because of a **weak functional sphincter + sharp angulation** at the sigmoid-rectum junction (~20 cm from the anus); a **mass movement** triggers the urge.
- **Internal sphincter = smooth muscle, involuntary; external sphincter = striated, voluntary via the pudendal nerve.** Defecation needs both relaxed.
- The **intrinsic myenteric reflex is weak alone**; the **parasympathetic sacral reflex via the pelvic nerves** makes it powerful, emptying from the **splenic flexure to the anus**.
- Deep breath + closed glottis + abdominal contraction + relaxed pelvic floor assist; **voluntary activation is less effective**, so habitual inhibition → **constipation**. Infants and cord transection empty automatically.
- **Peritoneointestinal** reflex → paralysis in peritonitis; **renointestinal / vesicointestinal** inhibit the gut.
Diarrhoea = ↑ motility + ↓ absorption; constipation = ↓ motility + ↑ water absorption.

Q Test yourself

1. As defecation proceeds, signals entering the spinal cord also act on the pelvic floor. Which response helps expel the feces?

- A. The pelvic floor is unaffected; only the abdominal wall muscles contribute
- B. The anal ring is pulled inward to compress the feces against the internal sphincter
- C. The pelvic floor relaxes downward while the anal ring is pulled inward
- D. The pelvic floor relaxes downward and the anal ring is pulled outward, evaginating the feces
- E. The pelvic floor contracts upward, tightening the anal ring around the feces

2. A 4-year-old with an infection that irritates the intestinal mucosa passes frequent watery stools and becomes dehydrated. Which mechanism best explains this?

- A. Intestinal paralysis from the peritoneointestinal reflex
- B. Failure of the parasympathetic defecation reflex, so the rectum never empties
- C. Loss of tonic constriction of the external anal sphincter, so stool dribbles constantly
- D. Excess water absorption in the colon because the contents stay too long
- E. Excessive motility moves contents through too quickly for absorption, so fluid and electrolytes are lost

3. A colorectal trainee asks why the rectum is usually empty. Which pair of features best explains this?

- A. Absorption of all fecal water in the rectum, so no bulk remains
- B. Tonic constriction of the external anal sphincter under pudendal control
- C. A weak functional sphincter and sharp angulation at the sigmoid-rectum junction
- D. Continual reflex emptying by the parasympathetic reflex several times an hour
- E. Retrograde peristalsis in the descending colon

4. A 52-year-old with acute peritonitis has stopped passing stool and has no bowel activity. Which mechanism is most likely responsible?

- A. The renointestinal reflex from irritation of the kidney
- B. The peritoneointestinal reflex, which inhibits excitatory enteric nerves and causes intestinal paralysis
- C. The parasympathetic defecation reflex overactivating the pelvic nerves
- D. Loss of voluntary control of the external anal sphincter
- E. Excess motility with decreased absorption

5. A surgeon regularly ignores the urge to defecate and instead relies on a deep breath and abdominal straining when it is convenient. Over time, which problem is she most likely to develop?

- A. Constipation, because reflexes started voluntarily are never as effective as natural ones
- B. Incontinence, because the pudendal nerve is damaged
- C. Diarrhoea, because the defecation reflexes become hyperactive
- D. Automatic emptying, as seen in newborns
- E. Intestinal paralysis, because of the peritoneointestinal reflex

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — D — Cord signals make the pelvic floor relax downward and pull outward on the anal ring to evaginate the feces.

2 — E — Diarrhoea = ↑ motility + ↓ absorption: chyme and feces pass too fast to be absorbed → dehydration and electrolyte imbalance.

3 — C — A weak functional sphincter ~20 cm from the anus plus sharp angulation at the sigmoid-rectum junction resist rectal filling.

4 — B — Peritoneal irritation → peritoneointestinal reflex → inhibited excitatory enteric nerves → intestinal paralysis (esp. peritonitis).

5 — A — Voluntarily initiated reflexes are less effective; people who often inhibit natural reflexes are likely to become constipated.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam